MAE 440 Section 01 Aerodynamics Laboratory

Experiment No.1

Shape Effects on Drag

Submitted By:

Toni Ensley

Maryam Fareed

Michelle Lee

Ravleen Maan

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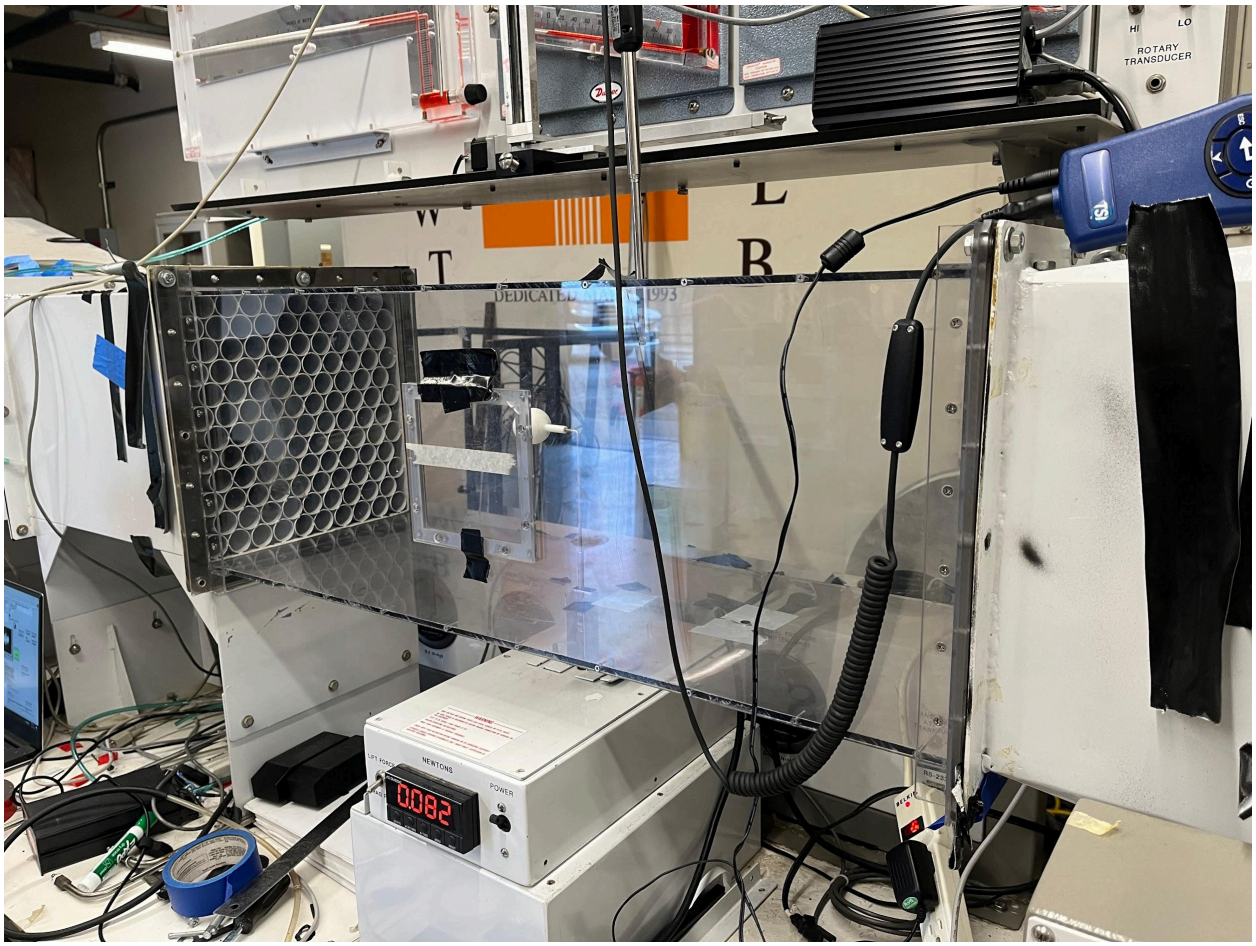
Abstract:

Drag is an aerodynamic force that counters an aircraft's forward motion through a solid or fluid but is heavily influenced by its shape. The purpose of this lab study is to analyze the effects of a shape's geometry when subjected to a low-speed wind tunnel environment. There were six shapes to analyze and all of which had a circular surface area: marshmallow, sphere, flat, mushroom, plunger, and a reverse cone. The experiment tested each shape at varying frequency references with a 10 Hz interval. An anemometer and a load cell were used to measure the velocity and drag force of each shape. With knowledge from previous courses and readings, streamlined bodies are assumed to be the more aerodynamic geometry because bluff bodies allow for flow to separate earlier. Despite this, the sphere has the lowest drag coefficient and Reynolds number at higher frequencies. The plunger was the least aerodynamic with the greatest drag coefficient. The marshmallow object consists of an abnormal drag coefficient at the lowest frequency reference but begins to drop and stabilize at higher frequencies.

Background/Theory:

Drag is a concept often studied in aerodynamics to improve the performance of objects in a fluid. A fluid flowing over an object starting at the leading edge will create a boundary layer that separates towards the back end. This separation can vary in size but it is what corresponds to the severity of drag force present. Drag force F_d is the combination of pressure drag and viscous drag. Pressure drag is due to an object's resistance to flow in a fluid, and is reliant on the shape of the object. In contrast, viscous drag is caused by skin friction and is reliant on its surface properties. Taking objects of various shapes but analyzing flow from a consistent reference area allows for the cross analysis of such properties. The relation between the drag coefficient $Cd = \frac{F_d}{\frac{1}{2}U_\infty^2 \rho A}$ and Reynolds value (Re) for each object when graphed gives a visualization of the various shape performances starting from low (laminar flow) to high (turbulent flow). Through the calculation of these factors for shapes such as a circle, cone, cylinder, and more it is possible to determine which shapes perform better in a dynamic environment.

Experimental Procedure:




D
L
B
W
T
Hampden
Hampden


ESC
VELOCICALC
20.06
1.844
19.6
76.8
Test 010


BALDOR
CATALOG NO. ID15H205-E
SERIAL NO. H050401008
SPEC NO. IN0965B00
INPUT 220 VAC 3 PH 50/60 HZ
OUTPUT 220 VAC 3 PH 0-400 HZ
OPERATING STANDARD 2.5 KHZ PWM QUIET 8 KHZ PWM
ZONE CONSTANT TORQUE* VARIABLE TORQUE CONSTANT TORQUE VARIABLE TORQUE
HP OUT 5 7.5 10 15
KW OUT 3.7 5.6 7.5 11.3
FLA OUT 18 27 37 55
FLA OUT* 16.5 25 33 50
KW IN 5.9 8.9 11.8 17.7
HP IN 7.9 11.9 15.8 23.6
Listed Industrial Control Equipment
Suitable for Use as Service Equipment
Open Type 1 Enclosure


Hampden®
MODEL H-6910-12-CDL
WIND TUNNEL


SPEED
LOAD
MAIN AC
BALDOR
FWD 192V 1476RPM
LOC 9.5A 50.00HZ
JOG
LOCAL
PROG
FWD
DISP
REV
SHIFT
ENTER
STOP
RESET
BALDOR
ADJUSTABLE SPEED DRIVE

The main objective was to test 6 different object shapes at 5 different speeds. To do this a wind tunnel test must be set up to record the speed (m/s) and the force (N) for each object. The same procedure is done for all 6 objects and is kept consistent to minimize any data recording errors. Setting up the windtunnel, a test stand inside the tunnel is used to hold the objects facing the incoming flow. Sticking a probe from the top of the tunnel above the object allows for the reading of velocity at that point. Turning the windtunnel on, the object has been placed on the test stand and the door to the windtunnel closed to seal the test section. Allow a couple seconds for the tunnel to adjust before taking any readings. To keep consistency with force being applied the increase was made by pressing the button 10 times for each reading taken. Each adjustment the velocity from the probe is recorded and this process done for a total of 5 readings. This exact process is repeated again after turning the airtunnel off and replacing the previous shape on the test stand with another.

Raw Data:

Flat Plate		**Marshmallow**		**Plunger**	
Dia =	**1.5”**	**Dia =**	**1.5”**	**Dia =**	**1.5”**

Table 1: Dimensions of sample items (flat plate, marshmellow, plunger)

Mushroom		**Sphere**		**Reverse Cone**	
Dia =	**1.5”**	**Dia =**	**1.5”**	**Dia =**	**1.5”**

Table 2: Dimensions of sample items (mushroom, sphere, reverse cone)

Frequency Reference (Hz)		**10**	**20**	**30**	**40**
Marshmallow	**F_d (N)**	0.035	0.021	0.049	0.081
	Velocity (m/s)	3.55	7.20	11.10	15.20
Sphere	**F_d (N)**	0.002	0.015	0.034	0.057
	Velocity (m/s)	3.62	7.25	11.25	15.35
Flat	**F_d (N)**	0.006	0.026	0.056	0.093
	Velocity (m/s)	3.52	7.26	11.31	15.43
Mushroom	**F_d (N)**	0.004	0.014	0.031	0.055
	Velocity (m/s)	3.57	7.30	11.15	15.30
Plunger	**F_d (N)**	0.007	0.028	0.064	0.110
	Velocity (m/s)	3.58	7.75	11.20	15.30
Reverse Cone	**F_d (N)**	0.005	0.026	0.055	0.096
	Velocity (m/s)	3.73	7.16	11.23	15.28

Table 3: Drag Force vs. Velocity for Different Shapes at Frequency References (Intervals of 10 Hz)

Sample Calculations and Results:

Drag coefficient formula: $C_D = \frac{F_D}{\frac{1}{2}\rho U_\infty^2 A}$

Reynolds number formula: $Re = \frac{\rho V D}{\mu}$

F_D = drag force

ρ = fluid density

U = V = free stream velocity

A = projected area of object

D = diameter

μ = dynamic viscosity

Example calculation for sphere model at 10 Hz.

Same logic is applied to all shapes and frequency references.

Assuming ρ_{air} is $1.225\ \frac{kg}{m^3}$

Converted diameter from 1.5 in to 0.038 m

Calculated area = $\frac{0.038^2 \pi}{4}$ = 0.0011395 m^2

$$C_D = \frac{0.002N}{\frac{1}{2}(1.204\frac{kg}{m^3})(3.62m/s)^2(0.011395m^2)} = 0.22236928182$$

$$Re = \frac{(1.204\ \frac{kg}{m^3})(3.62m/s)(0.038m)}{1.81x10^{-5}Pa*s} = 9174.50$$

Frequency Reference (Hz)	10	20	30	40	50
Marshmallow	4.048494	0.590523	0.579739	0.511070	0.446204
Sphere	0.222482	0.416004	0.391612	0.352647	0.309573
Flat	0.705908	0.719088	0.638183	0.569421	0.506753
Mushroom	0.457515	0.382970	0.363491	0.342501	0.314955
Plunger	0.796185	0.679574	0.743748	0.685002	0.625507
Reverse Cone	0.523883	0.739315	0.635748	0.599386	0.537192

Table 4: Calculated Drag Coefficient Values

Frequency Reference (Hz)	10	20	30	40	50
Marshmallow	8.97E+03	1.82E+04	2.81E+04	3.84E+04	5.00E+04
Sphere	9.15E+03	1.83E+04	2.84E+04	3.88E+04	4.94E+04
Flat	8.90E+03	1.84E+04	2.86E+04	3.90E+04	4.94E+04
Mushroom	9.02E+03	1.85E+04	2.82E+04	3.87E+04	4.95E+04
Plunger	9.05E+03	1.96E+04	2.83E+04	3.87E+04	4.94E+04
Reverse Cone	9.43E+03	1.81E+04	2.84E+04	3.86E+04	4.96E+04

Table 5: Calculated Reynolds Number

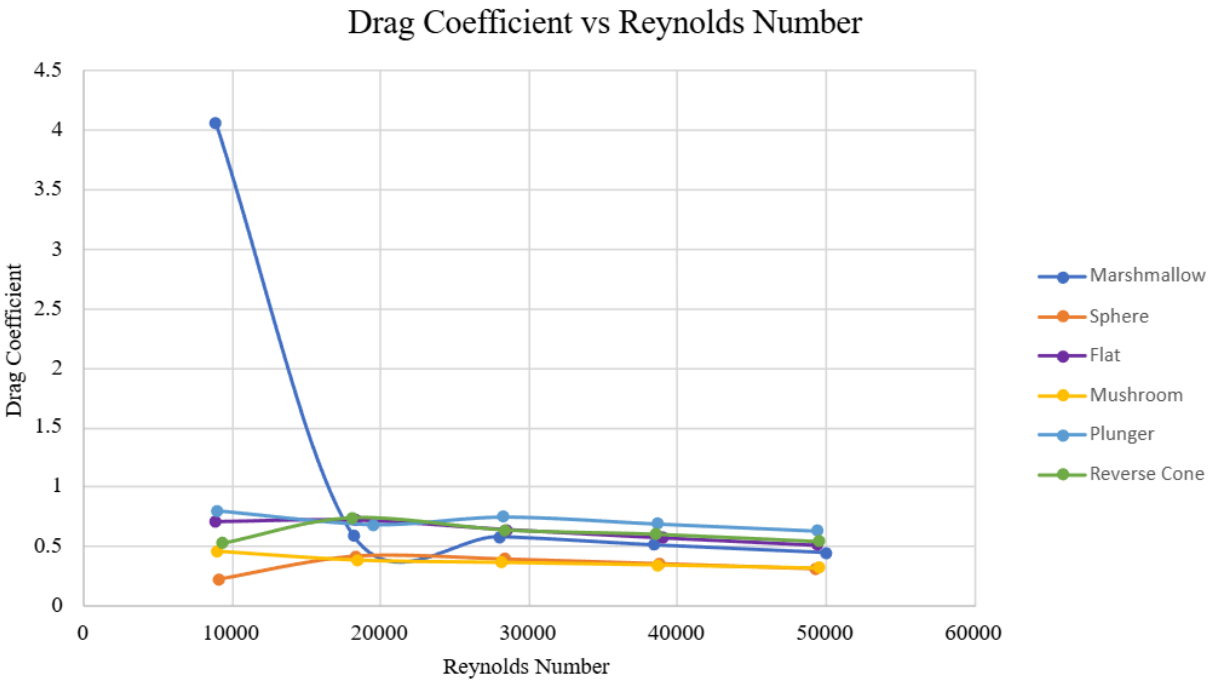


Figure 1: Drag Coefficient vs Reynolds Number

Discussion of Results:

The results demonstrate that the drag coefficient decreases as Reynolds number increases for the tested object shapes. This tells us the overall aerodynamic behavior of the object changed as we increased the flow frequency. The marshmallow object had a large drag coefficient of 4.05 at Reynolds number 8,970 and then dropped to 0.59 at Reynolds number 18,200 as referenced in Table 5: Calculated Reynolds Number. This result is a clear outlier, as seen in Figure 1: Drag Coefficient vs Reynolds Number, which may be due to error in measurement, the actual flow frequency, or the inconsistencies within the object itself. The overall drag coefficient of the mushroom ranged from 0.46 to 0.31. The sphere shows an overall consistent decrease starting from 0.22 and ending at 0.31, although there was another error between the readings increasing from 10 to 20 Hz. In comparison to the other shapes, the sphere holds a relatively low drag coefficient. On the other hand, the flat object had one of the higher drag coefficients ranging from 0.51 to 0.72. This data is accurate given the flat surface faced a large frontal area to the frequency flow, which generates a lot of flow. The plunger held the highest drag coefficient, reaching 0.80 at 10 Hz and dropping to only 0.74 at 30 Hz. The reverse cone reached its maximum value of 0.74 at 20 Hz, then decreased to 0.54 at 50 Hz. All these values are from Table 4: Calculated Drag Coefficient Values.

Overall, we can conclude that the flat shape and plunger produced the largest drag coefficient due to their geometric shape causing greater flow separation and presenting a larger area to the flow. On the other hand, the sphere and mushroom produced lower drag coefficients and showed more stable behavior. The marshmallow result at 10 Hz was significantly larger than all the other values and will be considered an outlier. At a Reynolds number above 18,000, the

drag coefficients for most shapes were the narrowest, indicating more consistent aerodynamic behavior at higher frequencies.

Conclusions and Recommendations:

This experiment analyzed the effects of geometry on drag force by testing six distinct three-dimensional shapes, under low pressure wind tunnel conditions. The total drag force experienced by a shape in low-speed aerodynamic conditions is a combination of pressure drag and viscous drag from surface friction. The experimental results confirmed the theoretical expectations that streamlined and rounded geometries yielded significantly lower drag coefficients C_D, whereas bluff bodies exhibited severe flow separation, leading to elevated pressure drag. Among the evaluated geometrical shapes, the sphere (C_D ranging from 0.22 to 0.42) and mushroom (C_D ranging from 0.31 to 0.46) demonstrated superior performance, closely aligning with the reference values for rounded 3-D bodies like spheres (C_D reaching 0.07-0.5). Conversely, non-streamlined shapes such as the flat plate (C_D reaching 0.72) and plunger (C_D reaching 0.80) caused early boundary layer separation and larger wake regions, driving up overall resistance. Plus, for Reynold number exceeding $1.8x10^4$, the drag coefficients for most shapes stabilized and began to drop, reflecting more consistent turbulent flow characteristics across higher velocity ranges.

To refine the accuracy of the experiment and reduce experimental error in future testing, addressing low-velocity data outliers and standardizing specimen rigidity are options. Resolving initial sensor stabilization lag and flow disturbances, such as the marshmallow's anomalous C_D value of 4.05 at 10 Hz, by allowing longer settling times prior to data capture and calibrating the load cell at low voltage outputs. Enforcing strict surface uniformity and rigid mounting on the test stand to eliminate structural deflection, micro-vibrations, or surface compression caused by

high dynamic pressure during elevated fan speeds is the second option that can be used to increase the accuracy of this experiment.

References:

Aerodynamics Laboratory Experiments, Written By Dr. Hamid R. Rahai & Dr. Jeremy Bonifacio

Group Contributions:

Michelle Lee - Abstract, Sample Calculations and Results

Toni Ensley - Background & Theory, Experimental Procedure

Maryam Fareed - Discussion of Results

Ravleen Maan - Conclusions and Recommendations